A05 ENHANCED AIR QUALITY | O (MAX : 4 PT)

Intent : Encourage and recognize buildings with enhanced levels of indoor air quality that promote the health and well-being of people.

Summary : This WELL feature requires projects to go beyond current guidelines to provide enhanced air quality levels that have been linked to improved human health and performance.[1,2]

Issue : The quality of the air people breathe indoors directly impacts their health and well-being and constitutes one of the most important aspects of buildings that can support human health. Researchers have also identified a clear relationship between indoor air quality and human productivity in buildings.[3] An average of 10% of productivity loss could be attributable to health issues related to poor indoor air quality in office buildings.[4] There is also an emerging body of evidence that air pollution can disrupt physical and cognitive development in children.[5] Studies have also shown that air pollution contributes to the large global burden of respiratory and allergic diseases, as well as the premature deaths of adults and children.[6] The premature mortality rate could be reduced by up to 15%, if PM_{10} is reduced from 70 to 20 μg/m^3.[7] As a result, enhanced air quality is positively correlated with improved health, cognitive and physical development, higher incomes and better economic performance.[1,2]

Solutions : Indoor air quality can be properly managed primarily through source control strategies, passive and active building design and operation strategies and human behavior intervention. High levels of indoor air quality require both professionals and building users to collaborate in the implementation of adequate approaches.

PART 1 MEET ENHANCED THRESHOLDS FOR PARTICULATE MATTER (MAX : 2 PT)

For All Spaces:

The following requirement is met:

a. Projects comply with the thresholds specified in the table below:

Tier	Particulate Matter Thresholds	Points
1	$PM_{2.5}$: 12 μg/m^3 or lower.[8] PM_{10}: 30 μg/m^3 or lower.[9]	1
2	$PM_{2.5}$: 10 μg/m^3 or lower.[9] PM_{10}: 20 μg/m^3 or lower.[9]	2

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET ENHANCED THRESHOLDS FOR ORGANIC GASES (MAX : 1 PT)

For All Spaces:

The following thresholds are met in occupiable spaces:

a.
   Acetaldehyde: 140 μg/m^3 or lower.[10]

b. One of the following:
   1. Acrylonitrile: 5 μg/m^3 or lower.[10]
   2. Caprolactam: 2.2 μg/m^3 or lower.[10]

c. Benzene: 3 μg/m^3 or lower.[10]

d. Formaldehyde: 9 μg/m^3 or lower.[10]

e. Naphthalene: 9 μg/m^3 or lower.[10]

f. Toluene: 300 μg/m^3 or lower.[10]

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 3 MEET ENHANCED THRESHOLDS FOR INORGANIC GASES (MAX : 1 PT)

For All Spaces:

The following thresholds are met:

a. Carbon monoxide: 7 mg/m^3 [6 ppm] or lower.[11]

b. Nitrogen dioxide: 40 μg/m^3 [21 ppb] or lower.[11]

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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